Literature Review: Autonomous Surgical Tool Handover with a Simulated 6-DoF Robotic Scrub Nurse

Selected application: Autonomous surgical tool handover using a simulated UR5e (6 degrees of freedom) as a robotic scrub nurse (RSN).

1. Background and Motivation

Scrub nurses hand instruments to surgeons, a repetitive task that needs constant attention. A global shortage of perioperative nursing staff exacerbates this workload and increases the risk of errors (Kang et al., 2026; Schäfer et al., 2023). Communication failures are also common in the operating room. Jacob et al. (2012) cite studies showing that about 31% of exchanges fail. RSNs aim to take over instrument handover, and the literature splits into four themes: interaction, gripping, handover quality, and software/simulation.

2. Interaction and Perception

Early RSNs such as Penelope and Quirobot relied on voice commands. Jacob et al. (2012) introduced **Gestonurse**, which recognizes static hand gestures by counting fingertips, with each finger count mapped to one of five instruments. It reached about 94.65% gesture accuracy and could pick instruments spaced at least 25 mm apart. However, it was 0.83 s slower than a human assistant (4.06 s vs. 3.23 s) and delivered to a fixed location, not to the surgeon's hand.

Kang et al. (2026) advanced this with **PERSN**, a pose-estimation-based RSN. It uses local voice recognition (Vosk), YOLO-based oriented bounding boxes to estimate 6D instrument pose, MediaPipe to estimate hand pose and handedness, and a grip-state classifier (99.2% accuracy). In a study with 17 surgeons, PERSN handed over 95% of instruments in the random receive-and-return task and 98% in the incision-and-suturing

task. Instruments were immediately usable 98% of the time, versus 82% for a predefined-location handover. Its handover time, however, was about 10.09 s compared with 2.37 s for a human nurse.

3. Gripping

Most RSNs use electromagnetic grippers (Penelope, Quirobot, Gestonurse), which only work on ferromagnetic tools. Two-finger grippers are the alternative (Kang et al., 2026). Schäfer et al. (2023) tested a granular jamming gripper, which molds around an object and locks under vacuum. It handled forceps reliably (10/10) but failed with flat instruments such as scalpels (0/10 successful transfers). No gripper handles every instrument well, so the gripper choice needs to be explicit in our design, or abstracted away in simulation.

4. Software Architecture and Simulation

Bonci et al. (2023) review **ROS2** (the second-generation Robot Operating System, a middleware for connecting robot hardware and software) and propose a modular cobot architecture. Its layers are perception, recognition, behavior planning, trajectory planning, re-planning, manipulation, and motion control. The authors show this working with MoveIt2 (the ROS2 motion-planning framework), an external RealSense depth camera, OpenCV, and OMPL's RRTConnect planner. They also warn that ROS2 real-time behavior is not guaranteed and depends on the developer. Their case study is a pick-and-place demo, not a surgical one.

For simulation, Yoon et al. (2023) compared the physics engines ODE, Bullet, Vortex, and MuJoCo. Bullet gave the best contact force and penetration behavior in simple scenarios and in peg-in-hole insertion. MuJoCo was fastest but showed large penetration and cannot handle nonconvex meshes. All engines failed the tight-tolerance bolting task. So contact-heavy interactions are the weak point of current simulators.

5. Research Gap

Existing RSN work is mostly hardware-specific, uses FANUC, KUKA, or Franka arms, and relies on fixed-position delivery or slow handovers. Bonci et al. give a reusable ROS2 architecture but no surgical use case. Yoon et al. show where simulation breaks down, which matters for handover contact. Nothing in these papers combines a modular ROS2 pipeline, a UR5e, and orientation-aware handover in simulation. That is the gap this project fills.

6. Robot Model Selection

Robots with 4–7 DoF were considered, including the 7-DoF Franka Panda and KUKA LBR Med14 used in the reviewed papers. We selected the **UR5e (6 DoF)**: it is a collaborative arm, has a mature ROS2 driver and MoveIt2 support, and needs no redundancy handling, so it is a good fit for simulated handover.

7. Proposed Draft Project Flow

The proposed project focuses on developing a simulated robotic scrub nurse using a UR5e robotic manipulator, ROS 2, and MuJoCo. The implementation will prioritize a functional autonomous surgical instrument handover pipeline. Advanced perception, manipulation, and motion-planning capabilities will be considered optional extensions depending on implementation progress.

1. **Simulation environment:** Develop a MuJoCo scene containing the UR5e manipulator, a simplified surgical instrument tray, one or more instruments, and a predefined surgeon-hand target. Simplified geometries will be used to reduce simulation complexity.
2. **Instrument localization:** Initially obtain instrument positions and orientations directly from the simulator. RGB-D camera simulation and computer-vision-based object pose estimation may be introduced as optional extensions.
3. **Command interface:** Implement a ROS 2 interface that accepts typed instrument requests and identifies the requested instrument. Voice recognition may be added if time permits.

4. **Behavior planning:** Develop a ROS 2 state machine that coordinates the handover process, including receiving a command, identifying the instrument, approaching it, grasping, delivering, releasing, and returning to standby.
5. **Grasping and handover:** Initially use a simplified simulated attachment mechanism rather than a fully modeled physical gripper. The robot will transport the instrument toward a predefined handover location while maintaining an appropriate orientation for transfer.
6. **Motion planning and control:** Start with predefined safe joint configurations and interpolated trajectories executed through the ROS 2–MuJoCo interface. Advanced inverse kinematics, MoveIt2-based planning, obstacle avoidance, and replanning may be added as extensions.
7. **Advanced interaction (optional):** Investigate dynamic surgeon-hand tracking, alternative instrument orientations, automatic instrument return, and tray reorganization.
8. **Evaluation:** Evaluate the simulated system using task completion rate, handover time, and, where measurable, positioning and orientation errors. Additional experiments may investigate different instrument positions or target configurations.

References

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